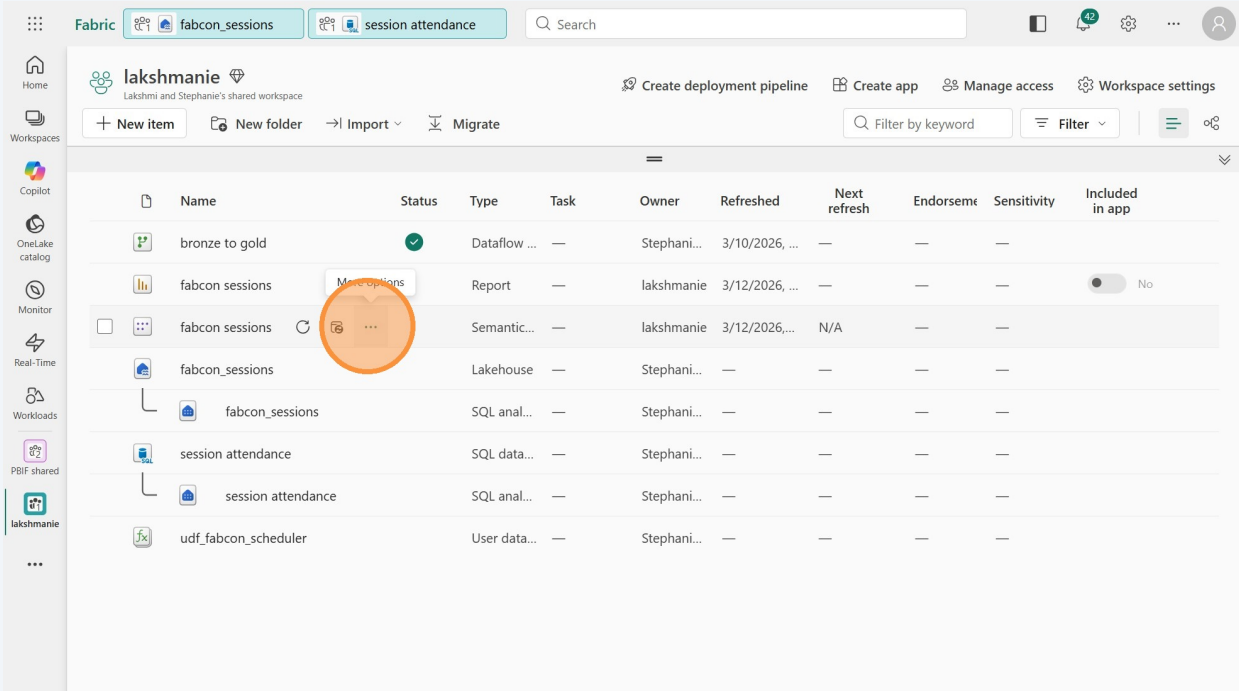


4b. Finish the semantic model in the service

Training.tips

We need to create relationships and measures before we can create the report.

- 1
- Go to your workspace if you're not already there. Hover over the name of the semantic model you just published and click the 3 dots to open the menu.



2

Click "Open semantic model." If you accidentally clicked on the name of the semantic model, there is a green "Open" button you can also click.

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
bronze to gold	✓	Dataflow ...	—	Stephani...	3/10/2026, ...	—	—	—	—
fabcon sessions		Report	—	lakshmanie	3/12/2026, ...	—	—	—	No
fabcon sessions		Semantic...	—	lakshmanie	3/12/2026, ...	N/A	—	—	—
fabcon_sessions				Stephani...	—	—	—	—	—
fabcon_sessions				Stephani...	—	—	—	—	—
session attendance				Stephani...	—	—	—	—	—
session attendance				Stephani...	—	—	—	—	—
udf_fabcon_scheduler				Stephani...	—	—	—	—	—

3

When you first open a semantic model, you're in "Viewing" mode. We need to be in "Editing" mode to make changes. Click the green "Viewing" dropdown in the top-right, and choose "Editing".

The interface shows the 'session attendance' and 'fabcon sessions' tabs. The 'Viewing' dropdown is highlighted, showing 'Editing' (Make any changes) and 'Viewing' (View, but make no changes). The main workspace displays a diagram with a 'gold speakers' table and its properties.

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
bronze to gold	✓	Dataflow ...	—	Stephani...	3/10/2026, ...	—	—	—	—
fabcon sessions		Report	—	lakshmanie	3/12/2026, ...	—	—	—	No
fabcon sessions		Semantic...	—	lakshmanie	3/12/2026, ...	N/A	—	—	—
fabcon_sessions				Stephani...	—	—	—	—	—
fabcon_sessions				Stephani...	—	—	—	—	—
session attendance				Stephani...	—	—	—	—	—
session attendance				Stephani...	—	—	—	—	—
udf_fabcon_scheduler				Stephani...	—	—	—	—	—

4

You will get a message that the model is being updated to the large semantic model storage format. Once it's finished, you can close the below message.

The screenshot shows the Microsoft Fabric interface. At the top, there are tabs for 'fabcon_sessions', 'session attendance', and 'fabcon sessions'. A green message bar states: 'This model was successfully converted to the large semantic model storage format. Version history is now available.' An orange circle highlights the 'Close' button on this message. To the right, the 'Properties' panel is open, showing settings for 'Cards'. The 'Show related fields when card is collapsed' toggle is set to 'Yes'. The main workspace displays a semantic model diagram with two tables: 'gold date' and 'gold speakers'. The 'gold date' table has columns: D, Date, Day_Name, Σ Day_of_Week, DDD, M, MMM, MMMYY, Σ Month, and a 'Collapse' link. The 'gold speakers' table has columns: Company, Description, Designation, picture_url, Speaker, speaker_id, Title, and a 'Collapse' link. A relationship line connects the two tables.

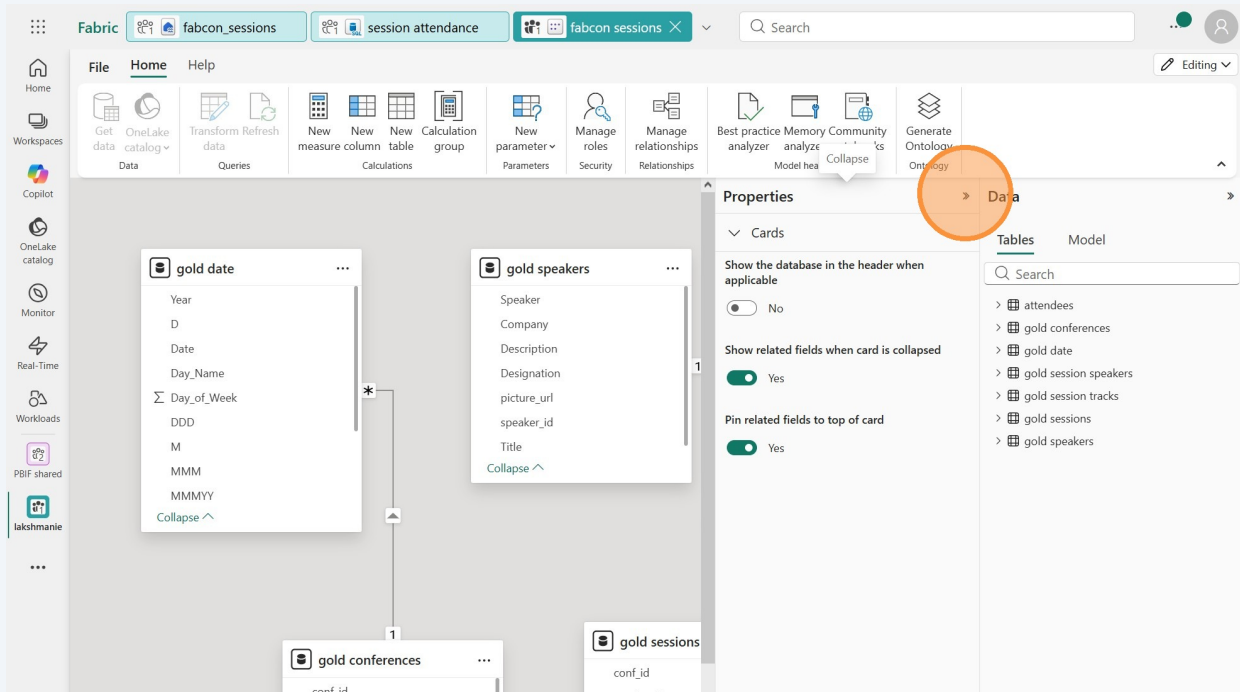
5

Optionally, click the toggle to "Pin related fields to top of card." Not required, but it can help to easily see which columns are in relationships.

This screenshot shows the same Microsoft Fabric interface as the previous one, but with the 'Pin related fields to top of card' toggle highlighted. The 'Properties' panel is still open, and the 'Show related fields when card is collapsed' toggle is set to 'Yes'. An orange circle highlights the 'Pin related fields to top of card' toggle, which is currently set to 'No'. The main workspace shows the same semantic model diagram with 'gold date', 'gold speakers', 'gold conferences', and 'gold sessions' tables. The 'gold sessions' table has columns: conf_id, conference, Date, Description, and End. A relationship line connects 'gold speakers' and 'gold sessions'.

6

If your screen is small, you can minimize the Properties pane to see more of the model.

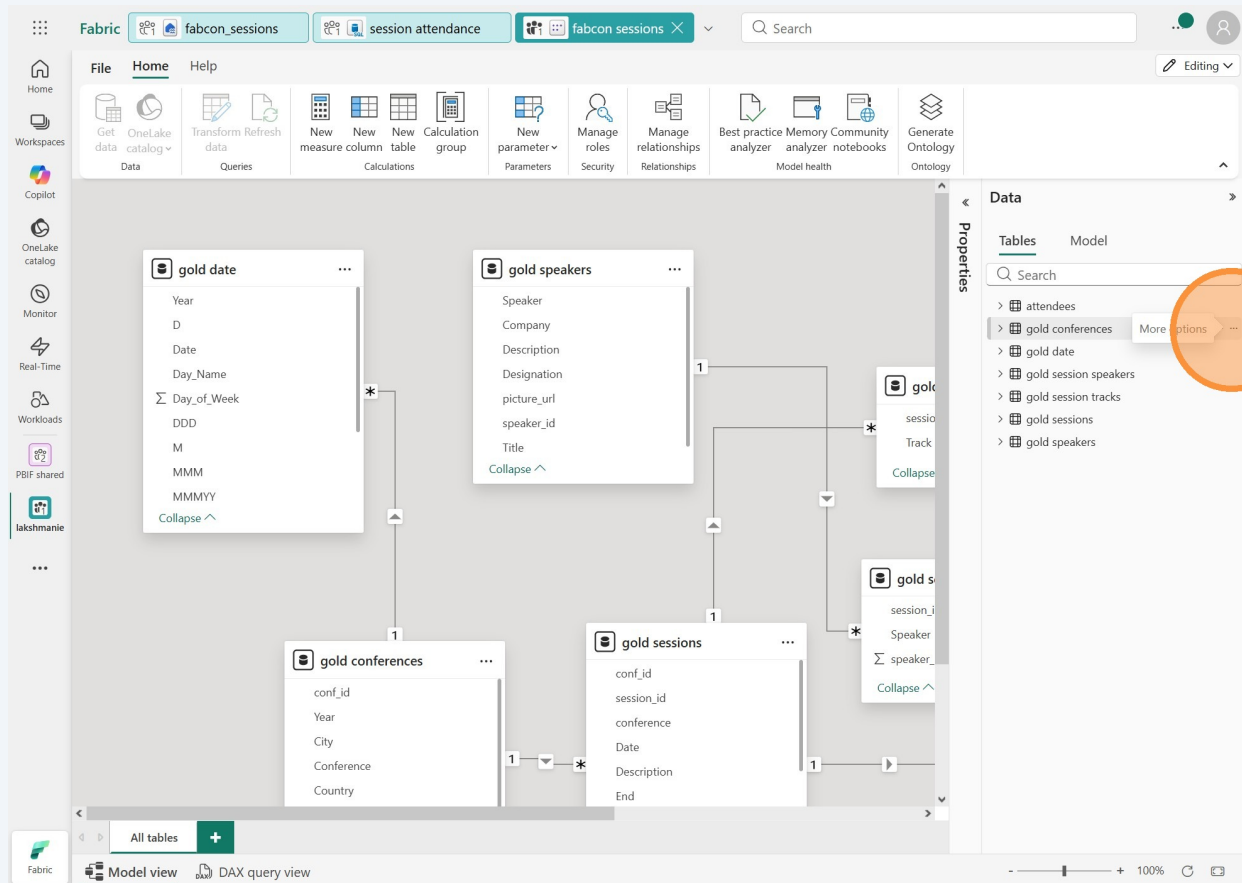


Rename the tables

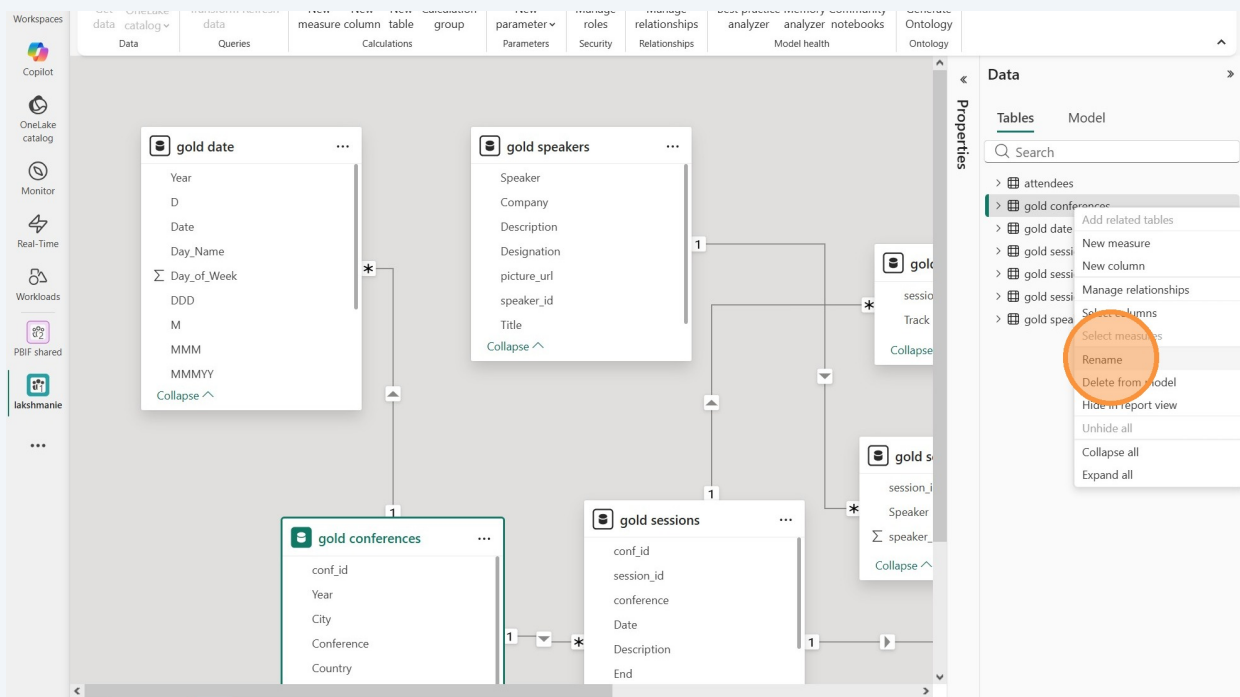
7

Depending on how you imported the tables, you may or may not have the word "gold" as a prefix on some of your tables. It's a good practice to have tables be named so that your colleagues can understand them easily. We're going to rename any of the tables that have "gold " in the beginning.

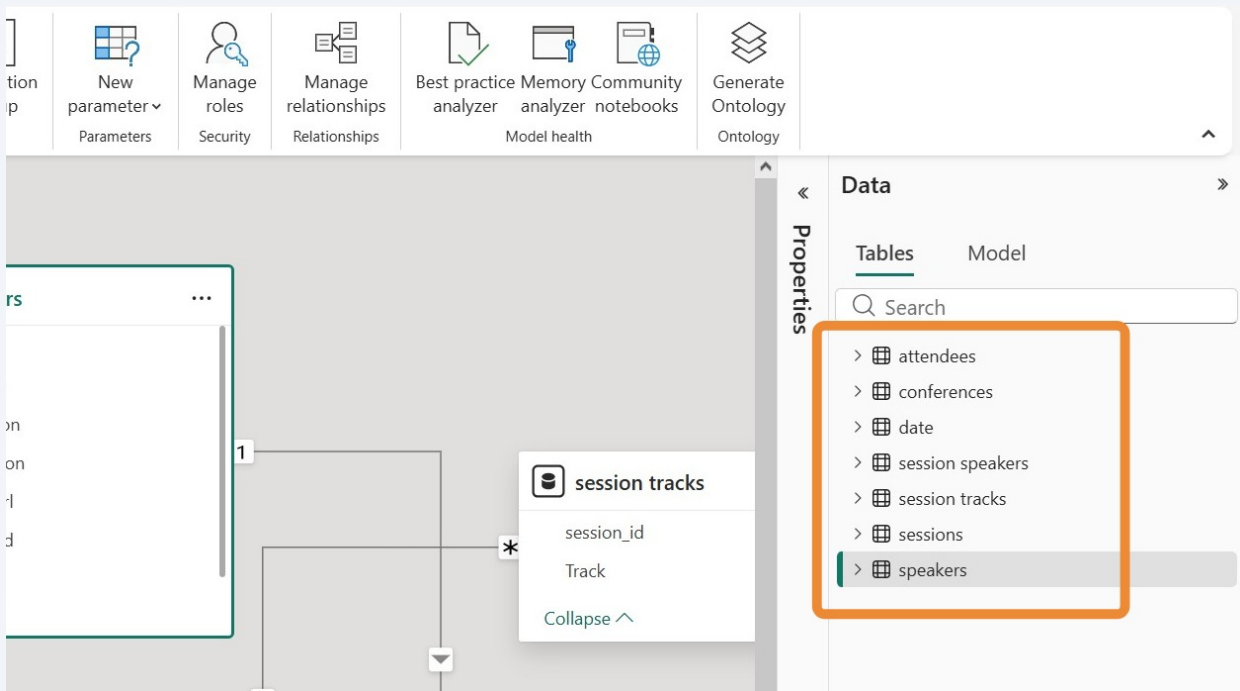
Look at your tables and find one that has "gold " in the beginning of the name. Hover over it and click the 3 dots.



8 Click "Rename" and delete "gold " from the beginning of the name.



9 Repeat the same process to any of your tables that have "gold " as a prefix. Once you're done your tables should be named like those below.



Add relationships

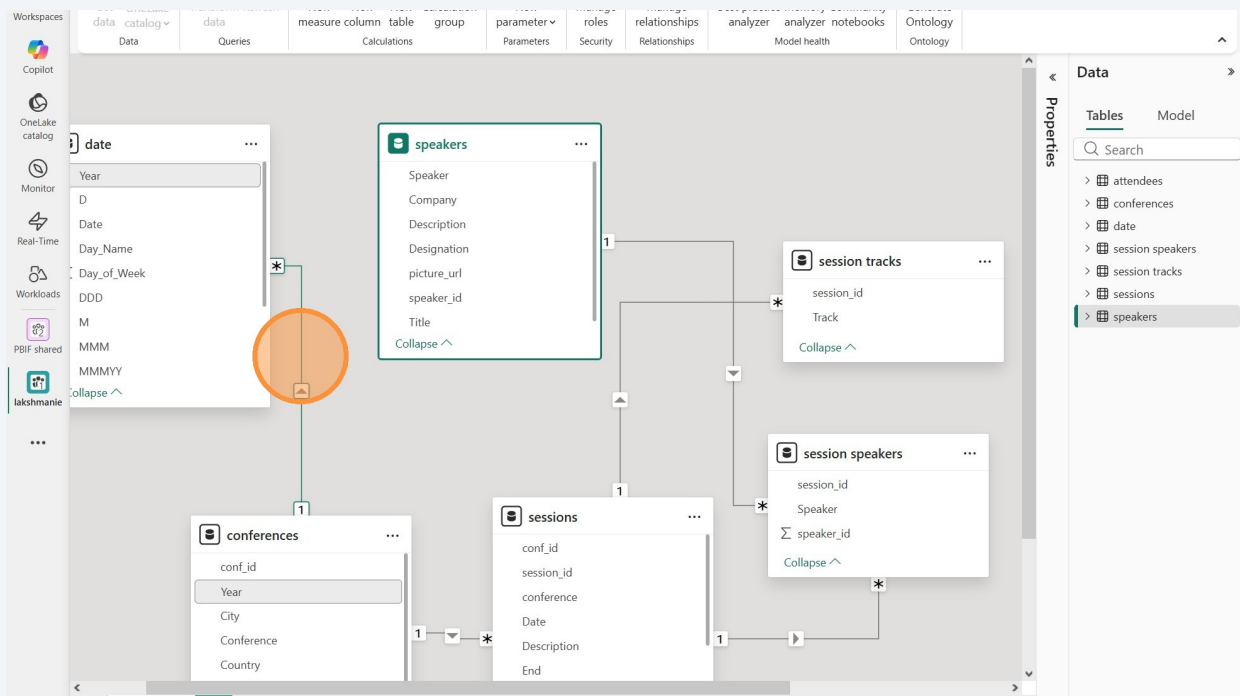
10

We need to edit the auto-created relationships and make sure our tables are properly related. You can rearrange the tables around so you can view them all.

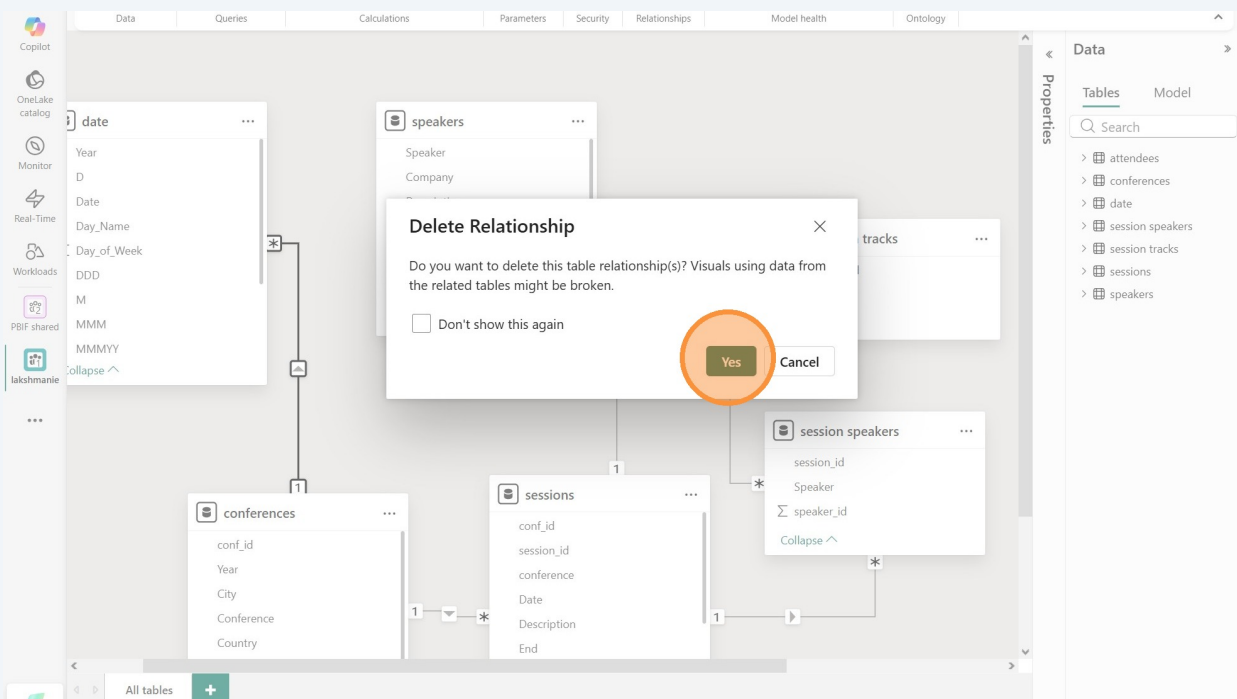
11

First, we don't need a relationship between the date and conferences table so let's delete that.

Right-click on the line connecting "date" to "conferences" and select "Delete."

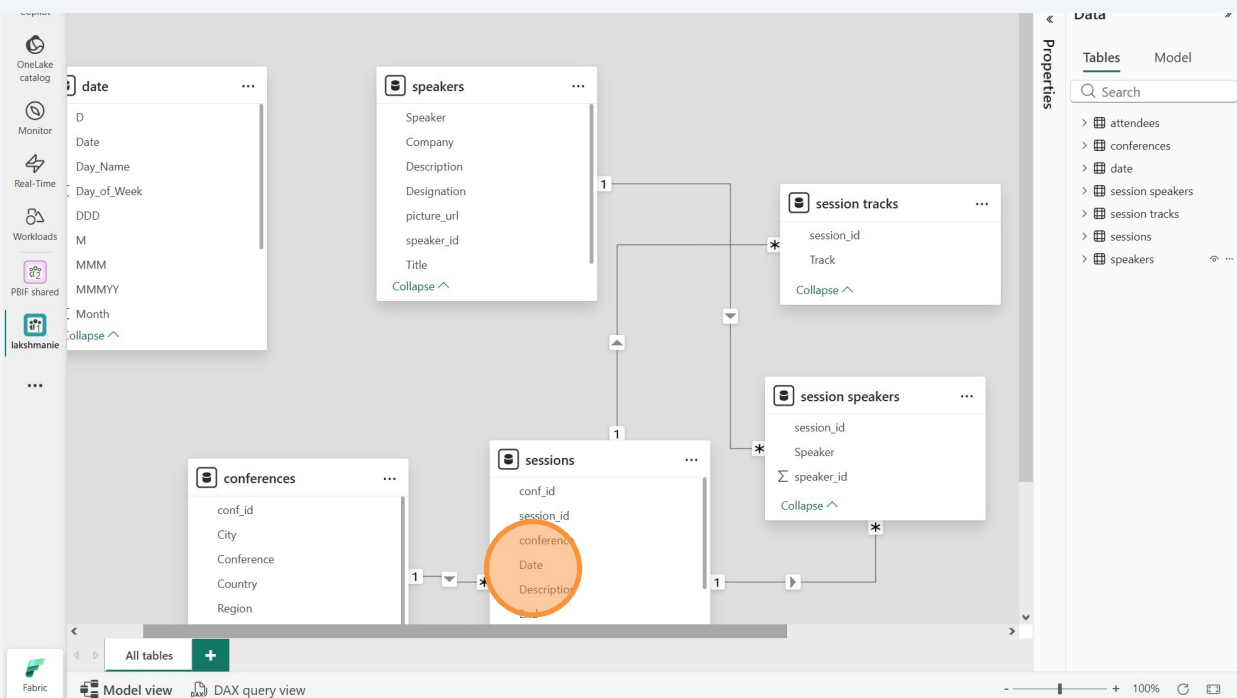


12 Click "Yes"



13 We do want to make a relationship between "session" and "date."

Click (and hold down the mouse button) the "Date" column from the "session" table and drag it to the "Date" column of the "date" table. This will open a relationship pop-up window.



14

The defaults here are correct. There can be one value in the date table with many rows in the session table. This makes sense because each date has multiple sessions. Click "Save."

The screenshot shows the Microsoft Fabric Data Catalog interface. On the left, a sidebar lists various tools like 'OnLake catalog', 'Monitor', 'Real-Time', 'Workloads', 'PBIF shared', and 'lakshmanie'. The main area displays the 'sessions' table with columns: conf_id, conference, Date, Description, End, enddt, and Length. Below this, the 'To table' section shows the 'date' table with columns: D, Date, Day_Name, Day_of_Week, DDD, M, and MMM. The 'Cardinality' is set to 'Many to one (*:1)' and the 'Cross-filter direction' is 'Single'. The 'Make this relationship active' checkbox is checked. The 'Save' button is highlighted with an orange circle.

conf_id	conference	Date	Description	End	enddt	Length
2000	FabCon2026U...	Thursday, Mar...	Get a behind ...	12/30/1899 9:...	3/19/2026 9:0...	1:00
2000	FabCon2026U...	Thursday, Mar...	Discover how ...	12/30/1899 9:...	3/19/2026 9:0...	1:00
2000	FabCon2026U...	Thursday, Mar...	Every great re...	12/30/1899 9:...	3/19/2026 9:0...	1:00

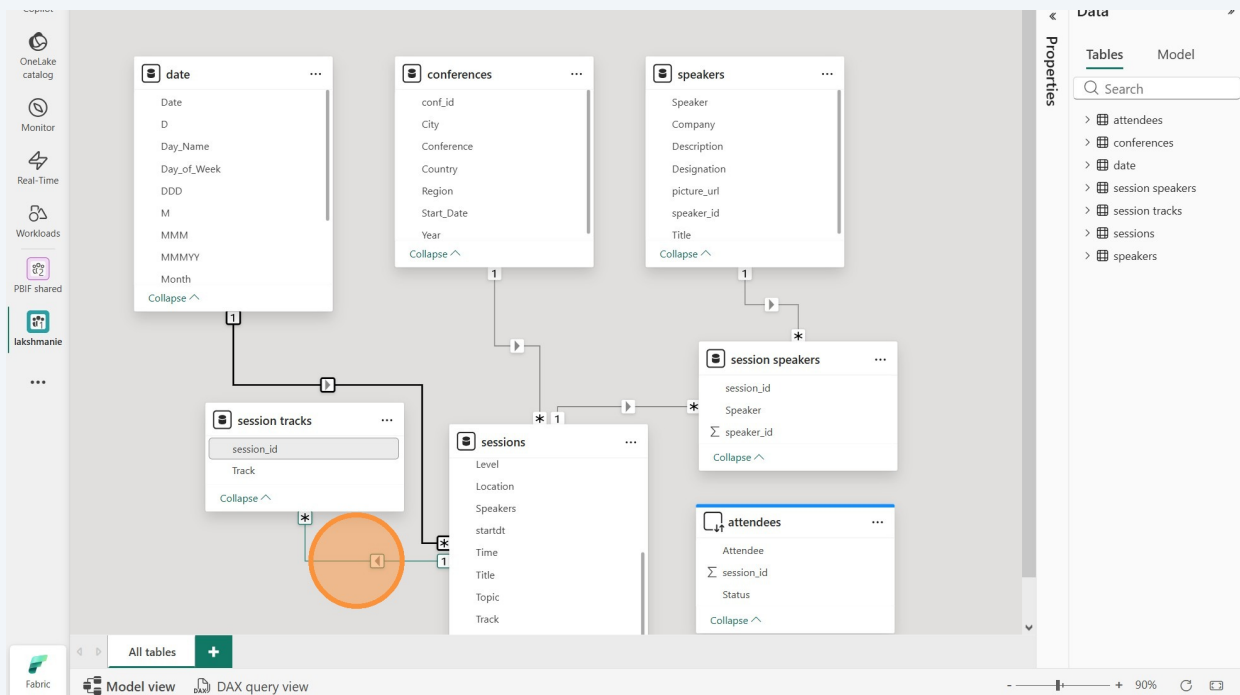
D	Date	Day_Name	Day_of_Week	DDD	M	MMM
T	1/1/2026	Thursday	4	Thu	J	Jan
F	1/2/2026	Friday	5	Fri	J	Jan
S	1/3/2026	Saturday	6	Sat	J	Jan

Cardinality: Many to one (*:1)
Cross-filter direction: Single
☒ Make this relationship active
☐ Assume referential integrity
☐ Apply security filter in both directions
Save Cancel

15

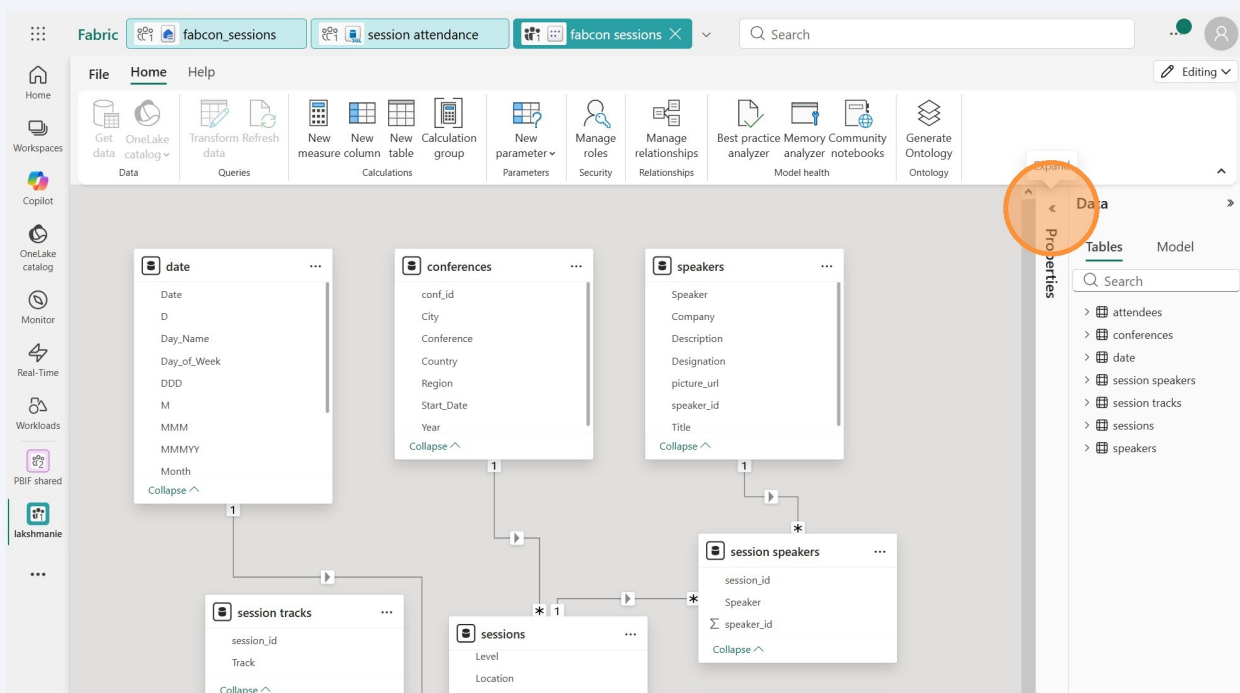
The automatically created relationship between sessions and session tracks is almost correct. Each session could have more than one track, and each track. So it does look correct! However, in our reports we would like to have the session tracks filter the sessions - not the other way around. So we want to change the directionality of that relationship.

Click on the line connecting sessions to session tracks.



16

If you've got the Properties pane collapsed, expand it.



17 Change the cross-filter direction from Single to Both.

The screenshot shows the Microsoft Fabric data model editor interface. On the left, a list of tables is visible: date, conferences, speakers, session tracks, sessions, session speakers, and attendees. The main workspace displays a diagram of these tables and their relationships. A relationship line connects the 'session tracks' table to the 'sessions' table. The 'Properties' pane on the right shows the relationship details. Under the 'Cross-filter direction' section, the 'Both' option is highlighted with an orange circle. The 'Apply changes' button is visible below the dropdown.

Properties

Relationship

Table: session tracks, Column: session_id

Cardinality: Many to one (*:1)

Table: sessions, Column: session_id

Make this relationship active: Yes

Cross-filter direction: Single (highlighted with orange circle)

Apply changes

Open relationship editor

18 Click "Apply changes"

The screenshot shows the same Microsoft Fabric data model editor interface as in the previous step. The 'Cross-filter direction' has been changed to 'Both'. An orange circle highlights the 'Apply change' button in the 'Properties' pane. The 'Apply changes' button is now labeled 'Apply change'.

Properties

Relationship

Table: session tracks, Column: session_id

Cardinality: Many to one (*:1)

Table: sessions, Column: session_id

Make this relationship active: Yes

Cross-filter direction: Both

Apply security filter in both directions: No

Apply change (highlighted with orange circle)

Open relationship editor

19

The "session speakers" table is what's sometimes referred to as a "bridge" table. We need to be able to have the speakers table filter the sessions those speakers are presenting. But if you look at the directionality of the relationships (which way the arrows are pointing), speakers properly filters "session speakers", but "session speakers" doesn't filter sessions. It's backward. Let's fix that.

Click on the line connecting sessions to "session speakers."

The screenshot shows the Microsoft Fabric data model editor interface. On the left, a sidebar contains navigation icons for OneLake catalog, Monitor, Real-Time, Workloads, and a user profile. The main workspace displays a data model with several tables: 'conferences', 'speakers', 'session speakers', 'sessions', 'session tracks', and 'attendees'. Each table is represented by a card showing its columns. The 'session speakers' table is highlighted with an orange circle, indicating it is the focus of the current action. The 'sessions' table is also highlighted. A relationship line connects the 'sessions' table to the 'session speakers' table. The 'Properties' pane on the right shows the relationship details for the connection between 'sessions' and 'session speakers'. The 'Table' dropdown is set to 'sessions' and the 'Column' dropdown is set to 'session_id'. The 'Cardinality' is set to 'Many to one (*:1)'. The 'Make this relationship active' toggle is turned on. The 'Cross-filter direction' is set to 'Both'. The 'Apply security filter in both directions' toggle is turned off. The 'Open relationship editor' button is visible at the bottom of the 'Properties' pane. The 'Data' pane on the right shows a search bar and a list of tables: attendees, conferences, date, session speakers, session tracks, sessions, and speakers.

20 Change the relationships Cross-filter direction from "Single" to "Both."

The screenshot shows the Microsoft Fabric Data Studio interface. On the left, a sidebar contains navigation icons for OneLake catalog, Monitor, Real-Time, Workloads, and a user profile. The main canvas displays a data model with the following tables and their columns:

- conferences**: conf_id, City, Conference, Country, Region, Start_Date, Year
- speakers**: Speaker, Company, Description, Designation, picture_url, speaker_id, Title
- session speakers**: session_id, Speaker, speaker_id
- sessions**: Level, Location, Speakers, startdt, Time, Title, Topic, Track
- attendees**: Attendee, session_id, Status
- session tracks**: session_id, Track

Relationships are shown as lines connecting tables. The 'session speakers' table is highlighted. The 'Properties' pane on the right shows the 'Cross-filter direction' dropdown set to 'Single'. An orange circle highlights the 'Both' option in the dropdown menu.

21 Click "Apply changes"

The screenshot shows the same data model as in step 20. The 'Cross-filter direction' dropdown in the 'Properties' pane is now set to 'Both'. An orange circle highlights the 'Apply changes' button in the 'Properties' pane.

22

The attendees table needs to be related to the sessions table. Grab the "session_id" column from the "attendees" table and drag it over to the "session_id" column of the "sessions" table.

The screenshot shows the Microsoft Fabric Data Studio interface. On the left, there's a sidebar with navigation options like 'OneLake catalog', 'Monitor', 'Real-Time', 'Workloads', 'PBIF shared', and 'lakshmanie'. The main workspace displays a data model with several tables: 'conferences', 'speakers', 'session speakers', 'sessions', 'attendees', and 'session tracks'. The 'attendees' table is highlighted with an orange circle. The 'sessions' table is also visible. The 'session_id' column in the 'attendees' table is selected. The 'Properties' pane on the right shows the 'Cards' tab with various settings like 'Show the database in the header when applicable', 'Show related fields when card is collapsed', and 'Pin related fields to top of card'. The 'Data' pane on the right shows a list of tables: 'attendees', 'conferences', 'date', 'session speakers', 'session tracks', 'sessions', and 'speakers'.

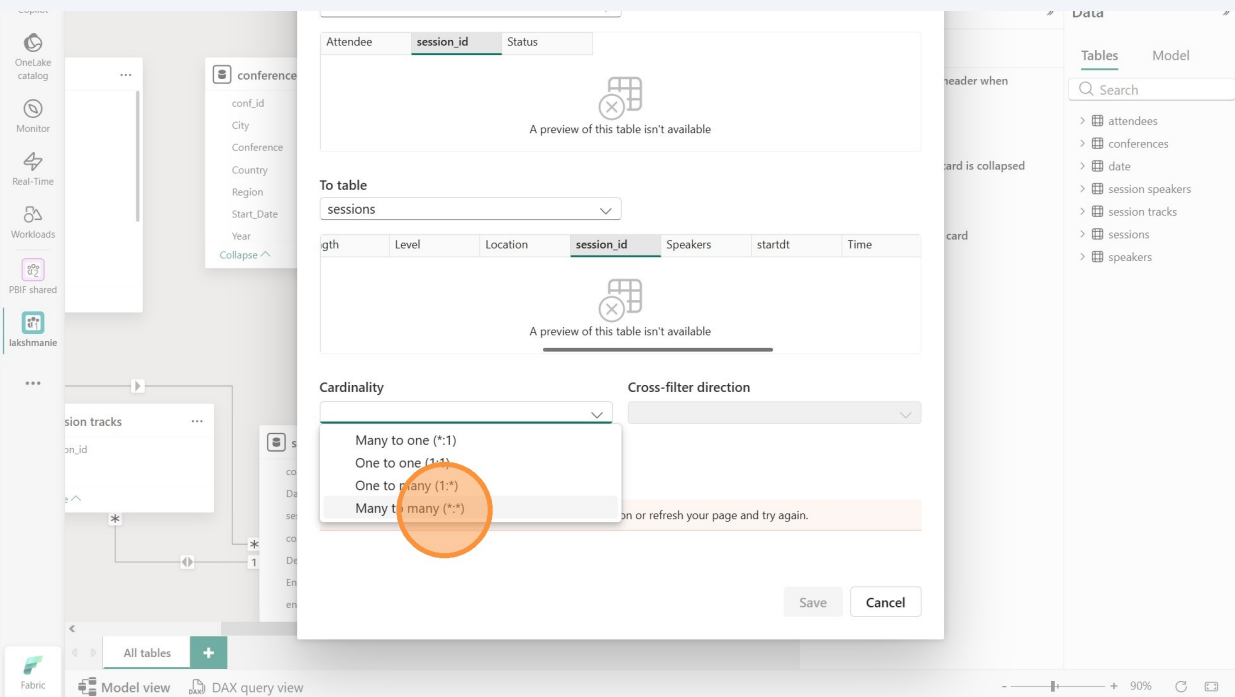
23

The "session_id" column of the attendees table is already selected. We just need to select the "session_id" column in the sessions table. Scroll to the right to find the session_id column. Click the session_id column to select it.

The screenshot shows the Microsoft Fabric Data Studio interface with a relationship dialog box open. The 'attendees' table is selected, and the 'session_id' column is highlighted. The 'sessions' table is selected as the 'To table'. The 'session_id' column is selected in the 'sessions' table. The dialog box shows a preview of the relationship and a warning message: 'Unable to validate this relationship. Select a new option or refresh your page and try again.' The 'Cardinality' and 'Cross-filter direction' are also visible in the dialog box.

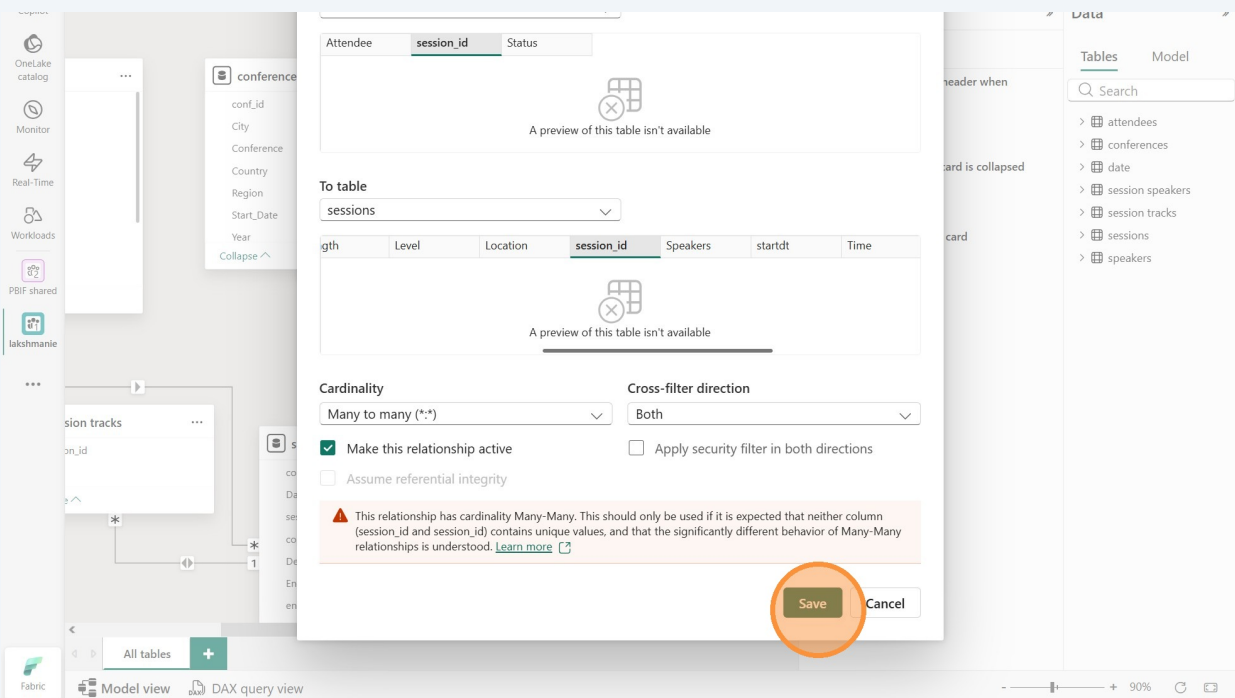
24

Click "Many to many" in the Cardinality dropdown. Each session can have multiple attendees and each attendee can attend multiple sessions.



25

Many-to-many relationships can be tricky, so you'll always get a friendly warning when you make them. It's cool. Click "Save."



26

You can minimize the Properties pane again now if you like. Let's make sure our relationships look good.

27

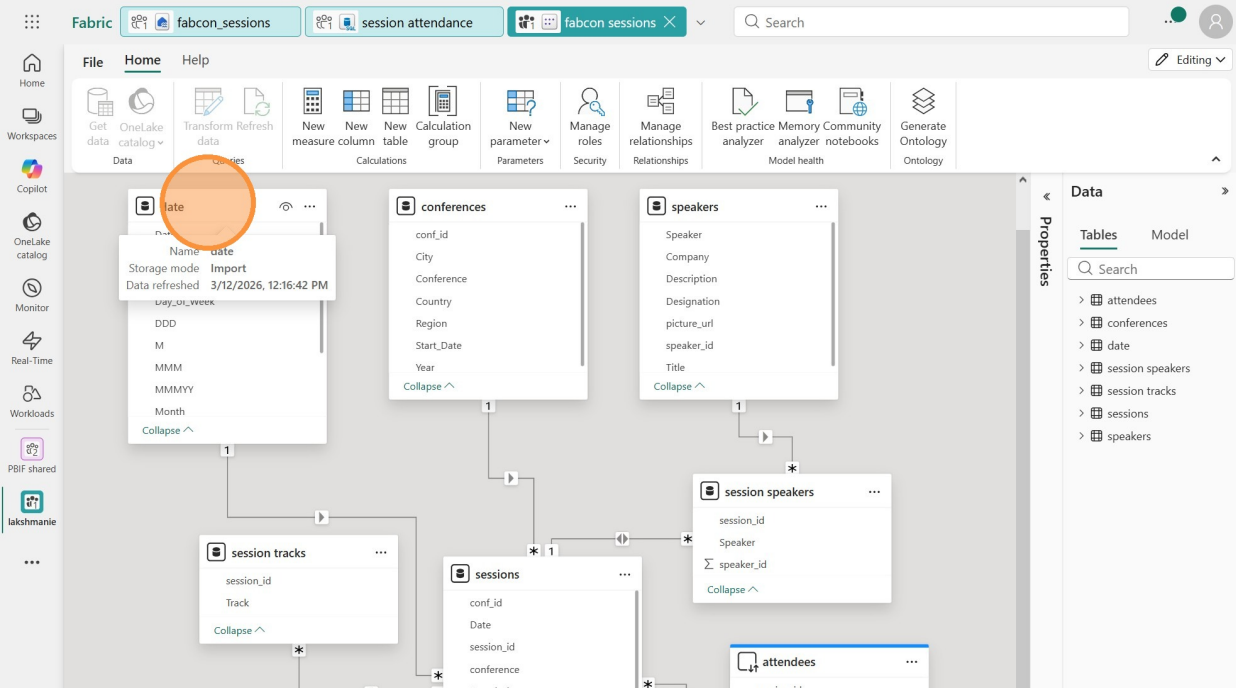
Take a minute to make sure that your relationships look like those below. You can rearrange your tables if it makes it easier.

Mark the Date table

28

To enable time intelligence calculations, it's always a good idea to have a date table and mark it as a date table. BUT - if you don't have a column in a table that is of the date datatype, that won't be possible. So first let's make sure we have a date column.

Click the date table to highlight it.



29 Click the "Date" column.

The screenshot shows the Microsoft Fabric data model editor interface. On the left, a vertical sidebar contains navigation icons for Home, Workspaces, Copilot, OneLake catalog, Monitor, Real-Time, Workloads, PBIF shared, and lakshmanie. The main workspace displays a relationship diagram with tables: 'date', 'conferences', 'speakers', 'session tracks', 'sessions', and 'attendees'. The 'date' table is highlighted with a green border. On the right, the 'Properties' pane is open, showing the 'General' tab. The 'Name' field is set to 'date', and the 'Description' field is empty. The 'Row label' dropdown is set to 'Select a row label'. The 'Key column' dropdown is set to 'Select a column with unique values'. The 'Is hidden' toggle is set to 'No', and the 'Is featured table' toggle is also set to 'No'. The 'Advanced' tab is partially visible at the bottom. In the background, a tooltip for the 'date' column is visible, showing the column name 'date' and its data type 'Date'.

30 The Date column is a "Text" data type. That doesn't sound right. Let's fix that.

The screenshot shows the Microsoft Fabric data model editor interface, similar to the previous one. The 'date' table is highlighted with a green border. On the right, the 'Properties' pane is open, showing the 'General' tab. The 'Name' field is set to 'Date', and the 'Description' field is empty. The 'Display folder' dropdown is set to 'Enter the display folder'. The 'Is hidden' toggle is set to 'No'. The 'Formatting' tab is selected, showing the 'Data type' dropdown set to 'Text'. The 'Format' dropdown is set to 'Text'. The 'Advanced' tab is partially visible at the bottom. In the background, a tooltip for the 'date' column is visible, showing the column name 'date' and its data type 'Text'.

31 Change it to the "Date" data type.

The screenshot shows the Microsoft Fabric data catalog interface. On the left, there's a sidebar with navigation options like Home, Workspaces, Copilot, OneLake catalog, Monitor, Real-Time, Workloads, and PBIF shared. The main area displays a data model with several tables: 'date', 'conferences', 'speakers', 'session tracks', and 'sessions'. The 'date' table is selected, and the 'Properties' pane on the right is open. In the 'Properties' pane, the 'Date' data type is highlighted in the dropdown menu. The dropdown options include Binary, True/false, Fixed decimal number, Date, Date/time, Decimal number, Text, Time, Whole number, and another Text option.

32 Click "Yes"

The screenshot shows the same Microsoft Fabric data catalog interface as before, but with a 'Data type change' dialog box open in the center. The dialog box contains the following text: 'With this data type change, your data will be stored differently. This may cause a loss of data or precision. After you make this change, you can restore the column by refreshing the table. Do you want to continue?'. There are two buttons at the bottom of the dialog: 'Yes' and 'Cancel'. The 'Yes' button is highlighted with an orange circle.

33

Now that the date table has a column that is of the date datatype, we can mark it as a date table. Click the 3 dots in the header of the date table.

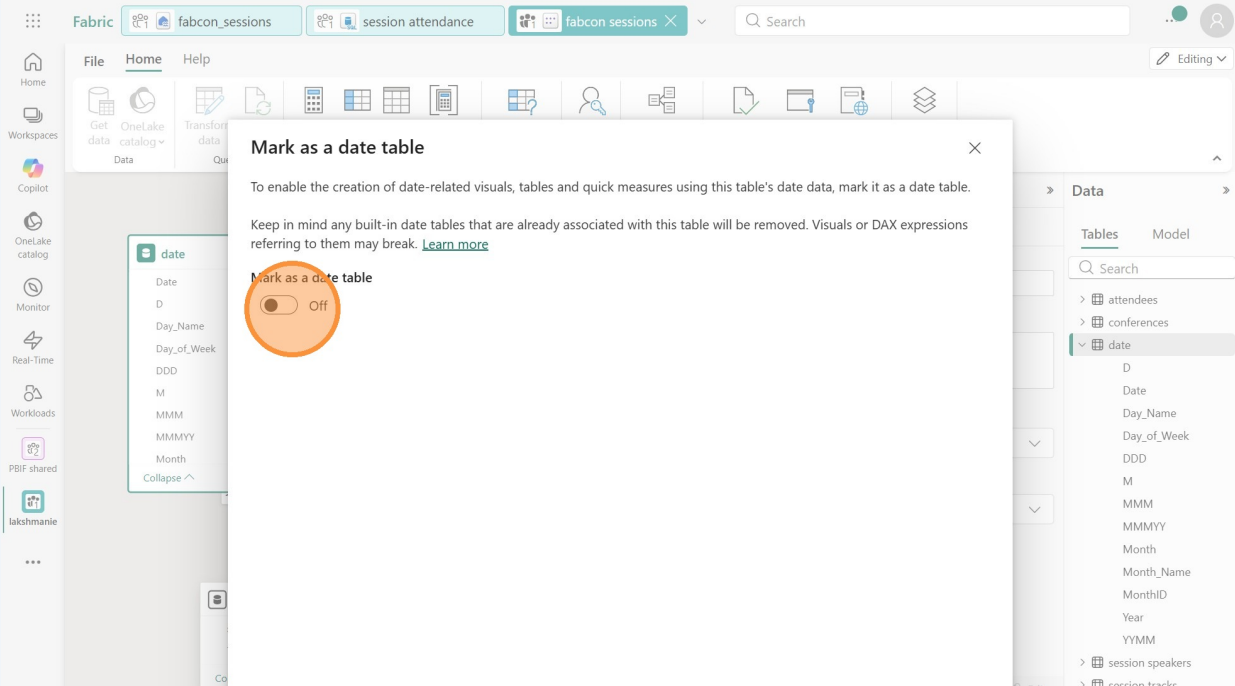
The screenshot shows the Microsoft Fabric interface with a data model. The 'date' table is highlighted, and its context menu is open, showing options like 'Mark as date table'. The 'sessions' table is also visible, and the 'date' table is connected to it. The 'session tracks' table is also visible, and the 'date' table is connected to it. The 'date' table's 'Date' column is highlighted with an orange circle, and its context menu is open, showing options like 'Mark as date table'.

34

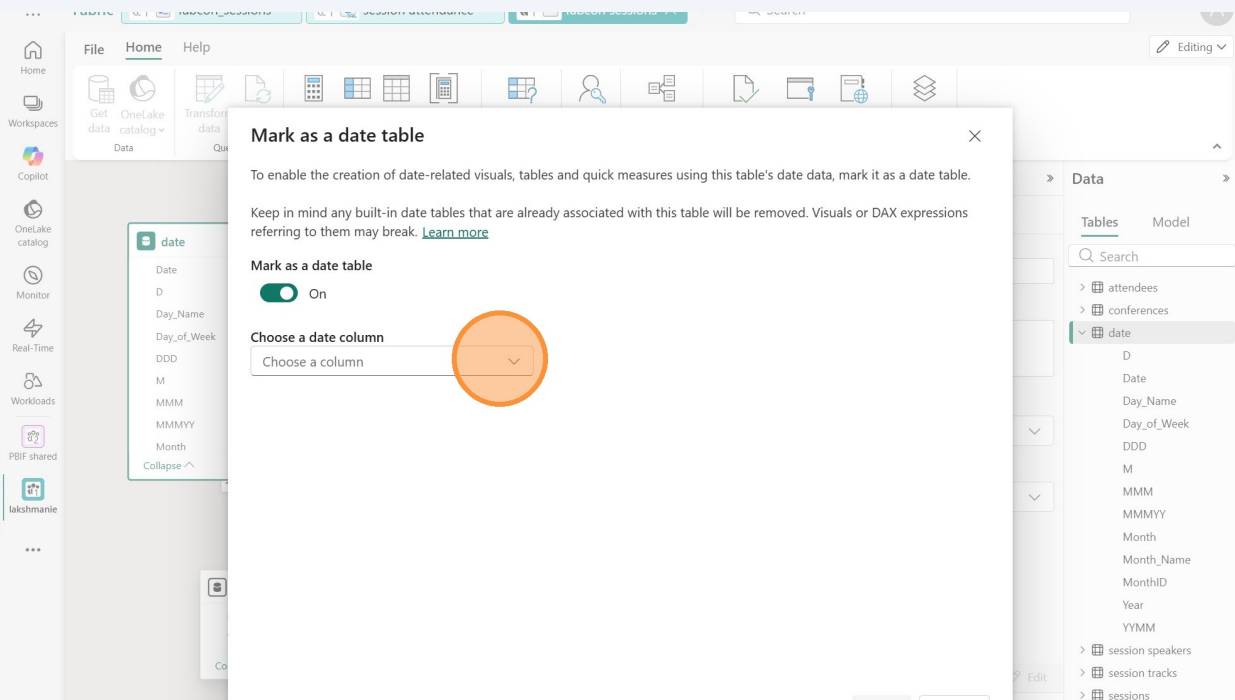
Click "Mark as date table"

The screenshot shows the Microsoft Fabric interface with a data model. The 'date' table is highlighted, and its context menu is open, showing options like 'Mark as date table'. The 'sessions' table is also visible, and the 'date' table is connected to it. The 'session tracks' table is also visible, and the 'date' table is connected to it. The 'date' table's 'Date' column is highlighted with an orange circle, and its context menu is open, showing options like 'Mark as date table'.

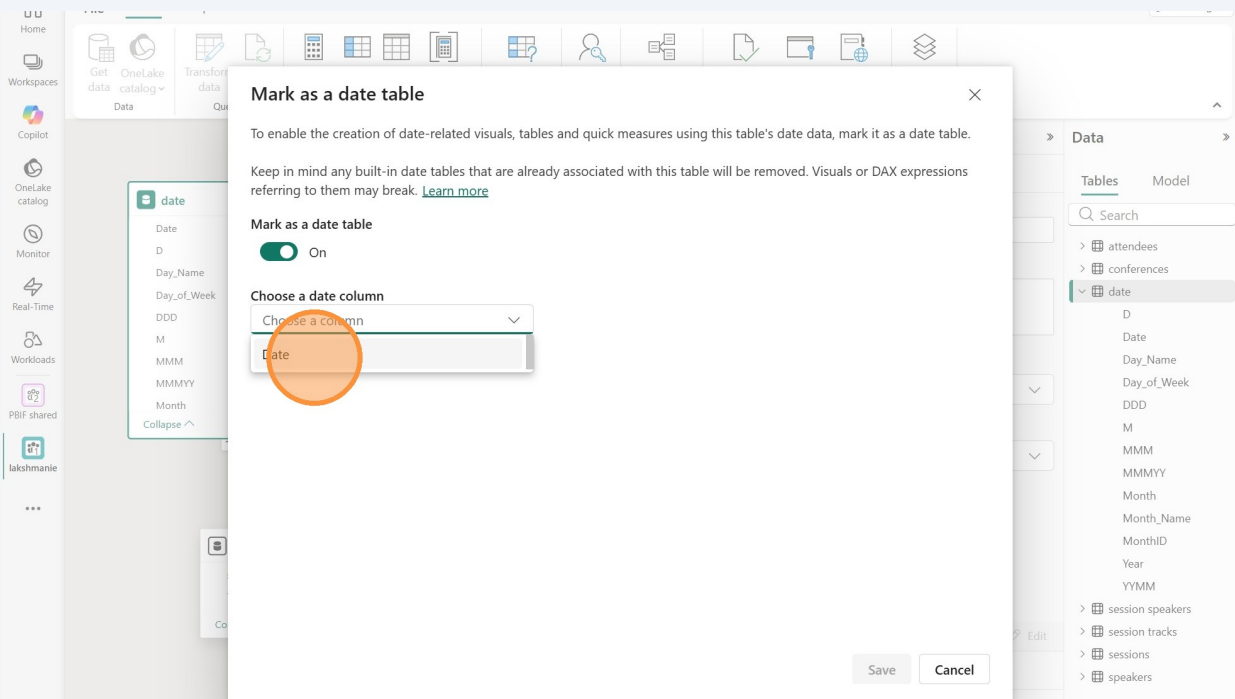
35 Turn the toggle switch to On.



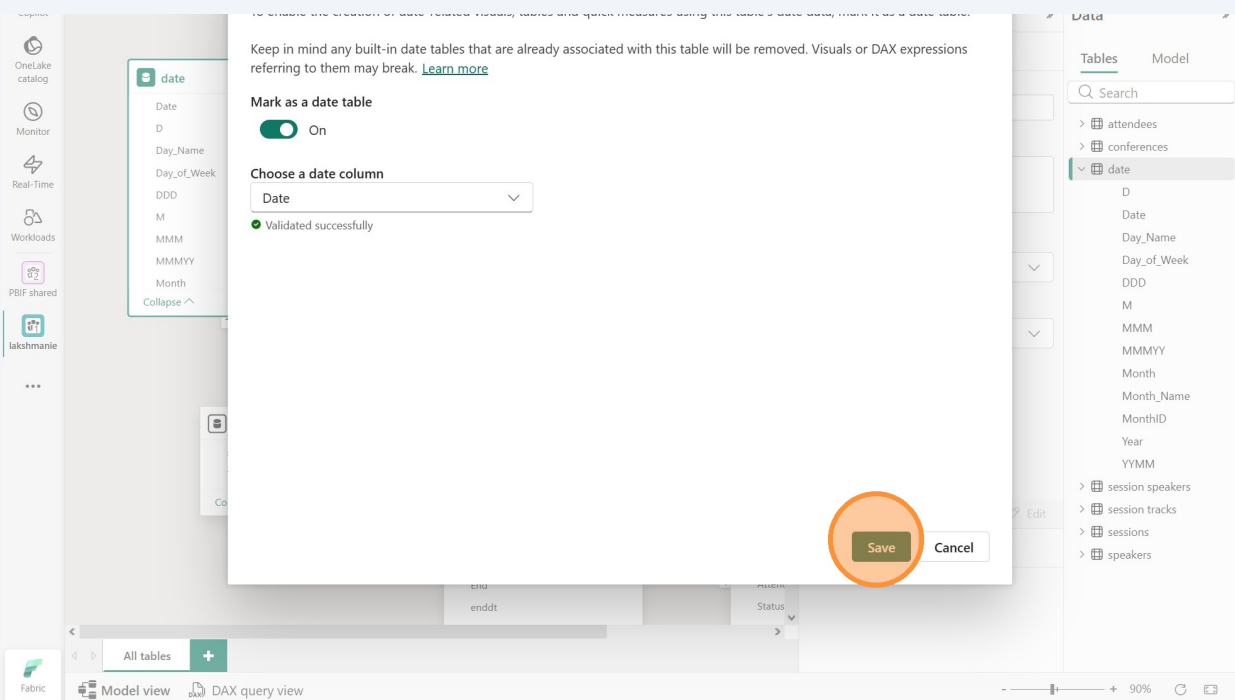
36 Click "Choose a column"



37 Click "Date"



38 Click "Save"

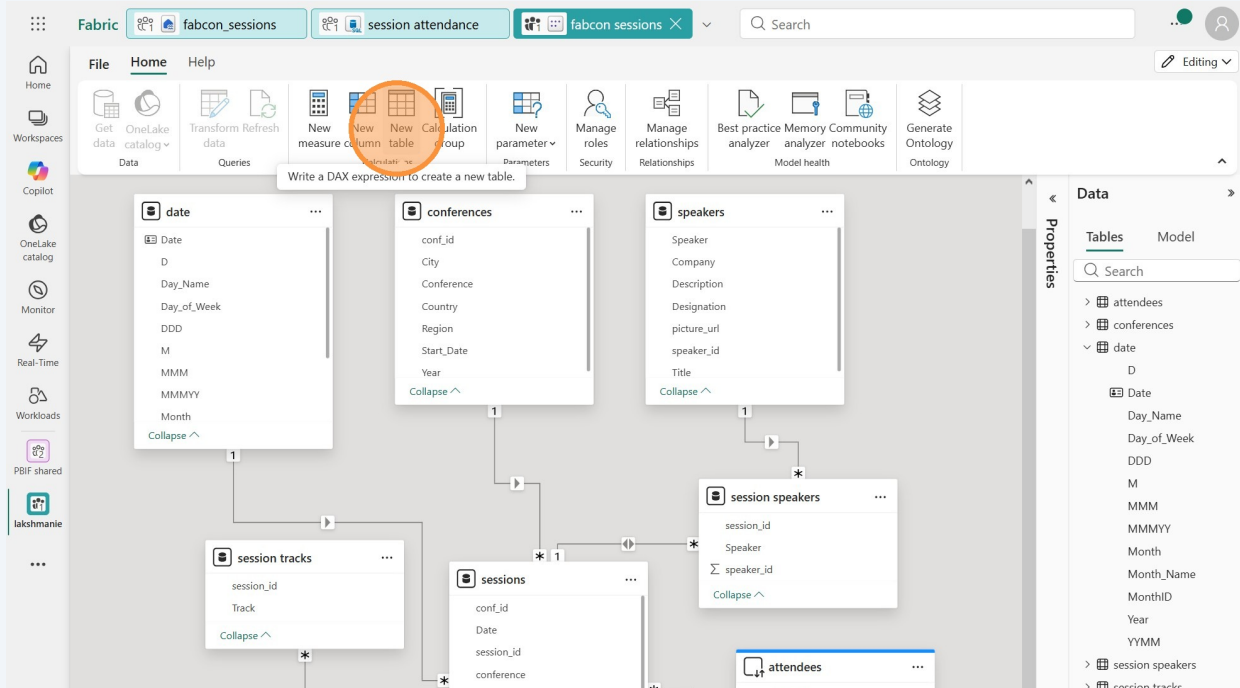


Create a table with DAX

39

We need one more table to populate a slicer for saving our session preferences. We want to be able to decide if we are "Definitely" or "Maybe" attending the sessions. So we'll create a table with just those two rows. No relationship will be required for this table.

Click the "New table" button in the Home tab.



40 In the formula bar, paste the following:

The screenshot shows the Microsoft Fabric interface. The formula bar at the top contains the following DAX expression:

```
1 attending status = DATATABLE (
2   "status", STRING,
3   {
4     { "Definitely" },
5     { "Maybe" }
6   }
7 )
```

The data model is visible in the background, showing tables like **session tracks**, **sessions**, **session speakers**, **attendees**, and **attending status**. The **attending status** table has a column named **status**.

41 Click the green checkmark to the left of the formula bar to enter it.

The screenshot shows the Microsoft Fabric interface. The formula bar at the top contains the following DAX expression:

```
1 attending status = DATATABLE (
2   "status", STRING,
3   {
4     { "Definitely" },
5     { "Maybe" }
6   }
7 )
```

A green checkmark is highlighted in the formula bar, indicating that the expression has been entered. The data model is visible in the background, showing tables like **sessions** and **attendees**.

Create a measures table

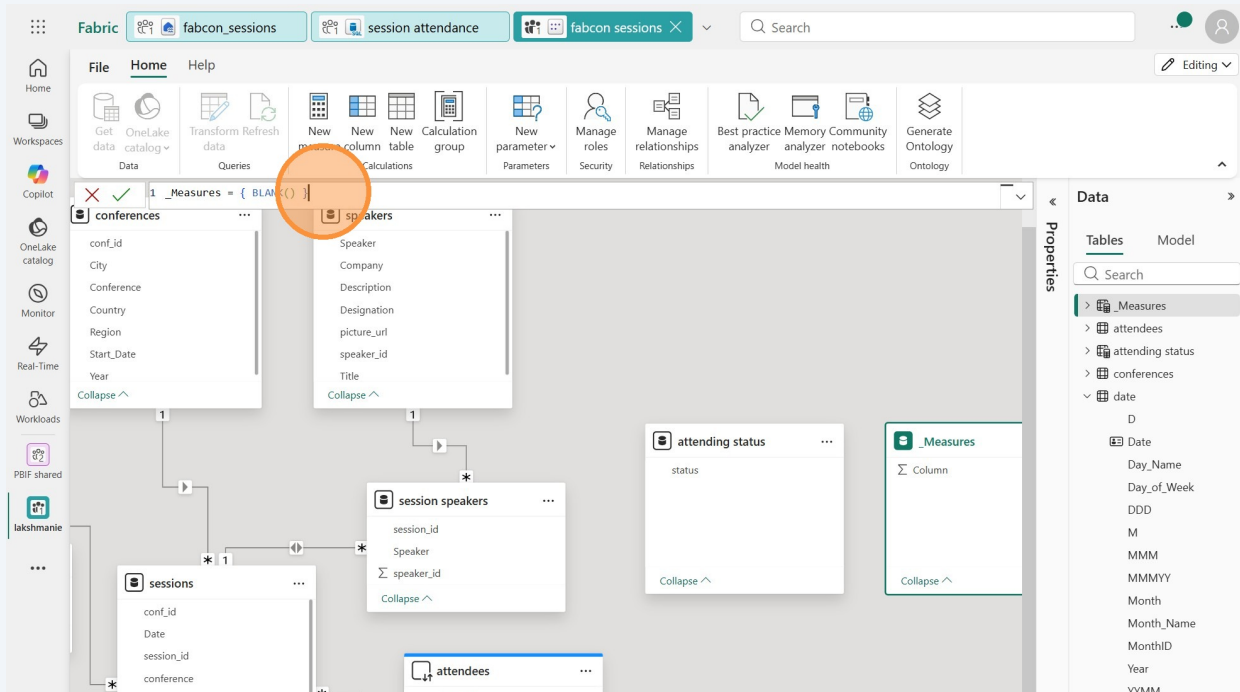


Tip! You don't have to create a special table for measures, and if you go down a rabbit hole of measure table research, you'll find many strong opinions about whether to put measures in their own table. Today we will, but that doesn't mean you have to. You can make your own choices! You do you.

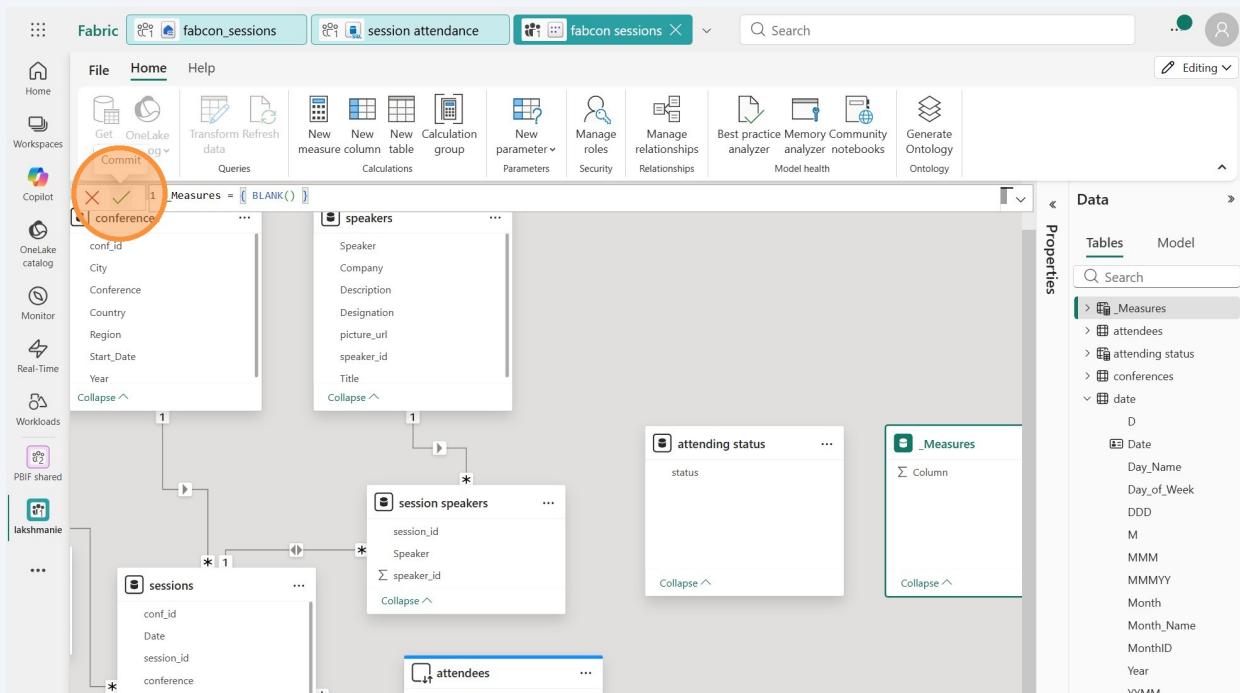
42 Click "New table"

The screenshot shows the Microsoft Fabric Data Studio interface. The top ribbon has tabs for 'File', 'Home', and 'Help'. Under the 'Home' tab, there are several buttons: 'Get data', 'OneLake catalog', 'Transform data', 'Refresh data', 'New measure', 'New column', 'New table', 'New calculation', 'New parameter', 'Manage roles', 'Manage relationships', 'Best practice analyzer', 'Memory analyzer', 'Community notebooks', and 'Generate Ontology'. The 'New table' button is highlighted with an orange circle. Below the ribbon, the main workspace shows a data model diagram with tables like 'date', 'session tracks', 'sessions', 'session speakers', 'attending status', and 'attendees'. The 'date' table is selected, and its properties are visible in the 'Properties' pane on the right. The 'Data' pane on the right shows a list of tables and their columns.

43 Enter "_Measures = { BLANK() }"

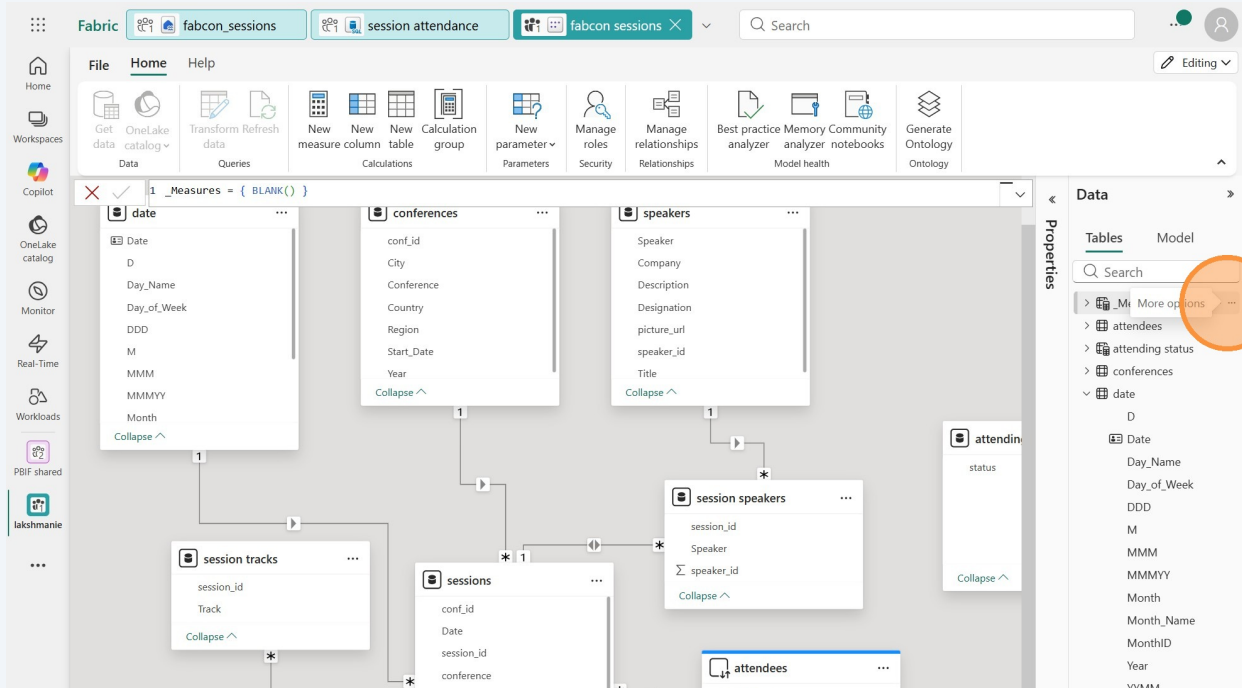


44 Click the green check mark to commit it.

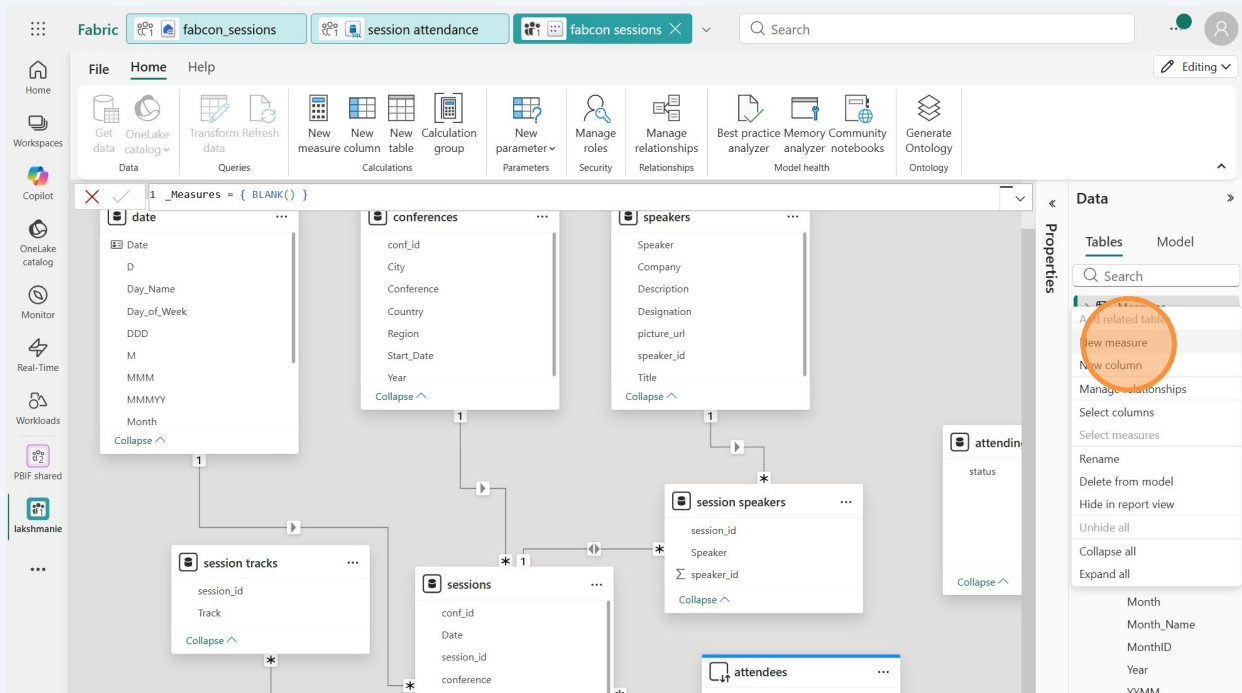


Create the measures

45 Click this the 3 dots next to the _Measures table.



46 Click "New measure"



47

Let's create a measure for session count. Replace the default text with:

Click the green check box or just click "Enter."

The screenshot shows the Microsoft Fabric Data Studio interface. The top navigation bar includes 'Fabric', 'fabcon_sessions', 'session attendance', and 'fabcon sessions'. The 'Home' tab is active, displaying a ribbon with various tools like 'Get data', 'Transform data', 'Queries', 'New measure', 'New column', 'New table', 'Calculation group', 'New parameter', 'Manage roles', 'Manage relationships', 'Best practice analyzer', 'Memory analyzer', 'Community notebooks', and 'Generate Ontology'. The main workspace shows a DAX measure named 'Session Count' with the formula: `Session Count = DISTINCTCOUNTNOBLANK(sessions[session_id])`. A green checkmark icon is visible next to the formula bar, indicating the measure is ready to be saved. The 'Data' pane on the right shows a list of tables and measures, including 'sessions', 'speakers', 'session speakers', 'attending status', and '_Measures'. The 'sessions' table is expanded, showing columns: 'conf_id', 'Date', 'session_id', and 'conference'. The 'speakers' table is also expanded, showing columns: 'Speaker', 'Company', 'Description', 'Designation', 'picture_url', 'speaker_id', and 'Title'. The 'session speakers' table is expanded, showing columns: 'session_id', 'Speaker', and 'speaker_id'. The 'attending status' table is expanded, showing the 'status' column. The '_Measures' table is expanded, showing the 'Measure' column. The 'Data' pane on the right also shows a list of tables and measures, including 'sessions', 'speakers', 'session speakers', 'attending status', and '_Measures'.

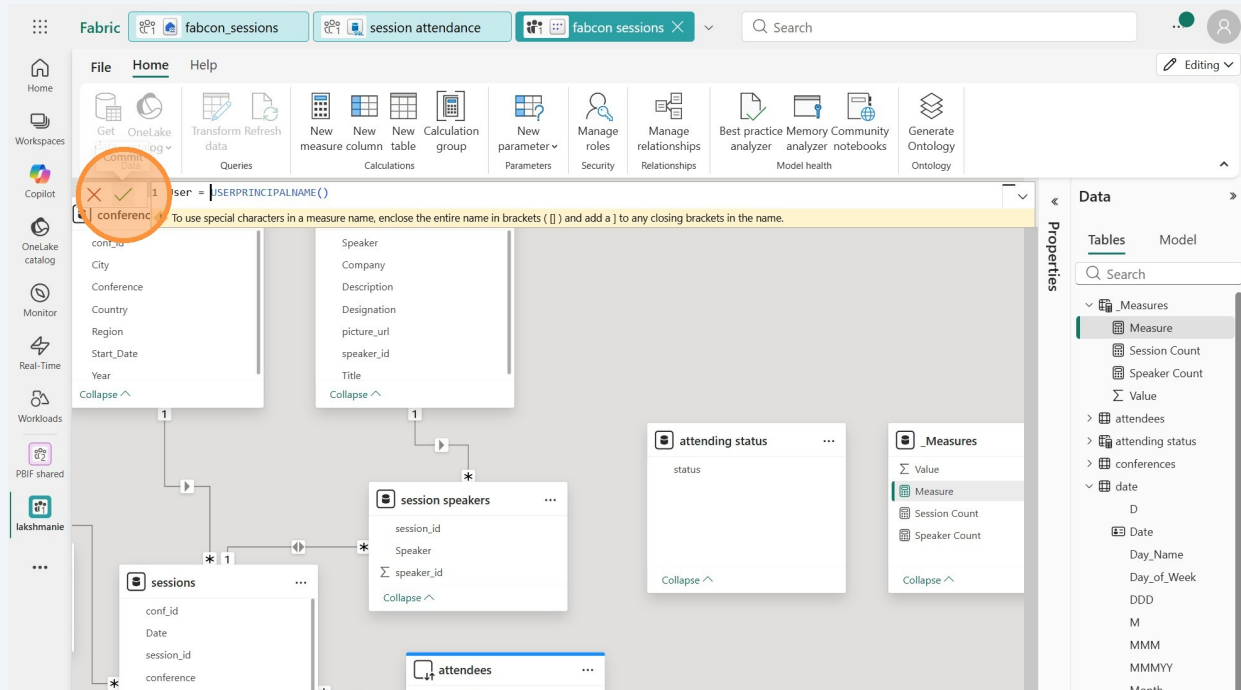
48

Great job! Let's make another measure! Repeat the last steps to create another measure in the `_Measures` table.

Replace the default text with the following measure:

User = USERPRINCIPALNAME()

This measure will return the UPN of the user running the report. We'll use it for the attendees updates.



Tip! In the measures you create below - ensure you're referring to the same table and column names that you created in your lakehouse and your SQL database. If you get a red squiggly on the `(table[(columnname)])` - update those with the correct table and column names that you have in your data!

49

Do the same as above to create the following four measures:

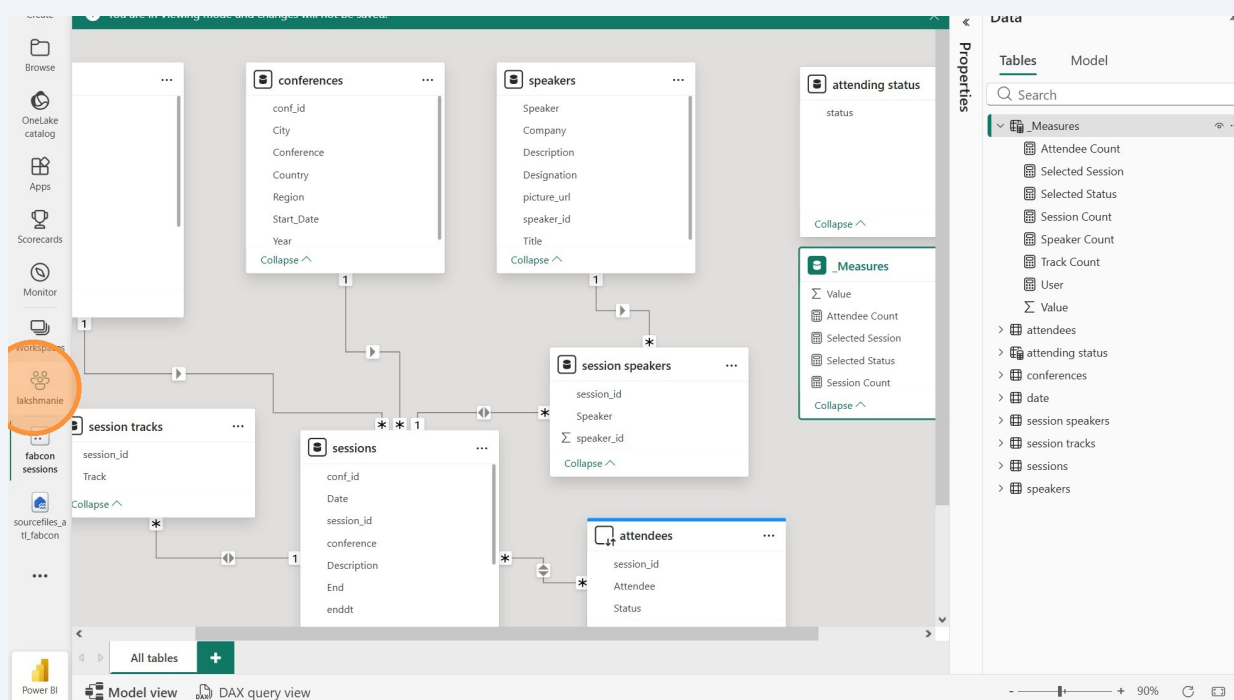
```
[[Selected Session = SELECTEDVALUE(sessions[session_id]]]
[[Selected Status = SELECTEDVALUE('attending status'[status]]]
[[Speaker count = DISTINCTCOUNTNOBLANK(speakers[Speaker]]]
[[Attendee Count = DISTINCTCOUNTNOBLANK(attendees[Attendee]]]

[[Track Count = DISTINCTCOUNT('session tracks'[Track]]]
```

We'll use the selected session and selected status measures for the User Defined Function we created earlier.

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Make sure your model has the below relationships and measures. Then go back to your workspace.



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Click your semantic model to view info about it.

POWER BI | lakshmanie

Home | Create deployment pipeline | Create app | Manage access | Workspace settings

+ New item | New folder | Import | Migrate

Filter by keyword | Filter

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
bronze to gold	✓	Dataflow ...	—	Stephani...	3/15/2026, ...	—	—	—	
copyjobgold		Copy job	—	Stephani...	—	—	—	—	
fabcon sessions		Report	—	lakshmanie	3/12/2026, ...	—	—	—	No
fabcon sessions		Semantic...	—	lakshmanie	3/12/2026, ...	N/A	—	—	
fabcon_sessions		Lakehouse	—	Stephani...	—	—	—	—	
session attendance		SQL data...	—	Stephani...	—	—	—	—	
session attendance		SQL anal...	—	Stephani...	—	—	—	—	
udf_fabcon_scheduler		User data...	—	Stephani...	—	—	—	—	

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If you expand the `_Measures` table you can see the measures you created. You can scroll through all the tables.

Open | AI summary (preview) | File | Refresh | Prep data for AI | Explore | Write DAX queries

Semantic model
fabcon sessions

Overview | Lineage | Monitor | Permissions

Make this item more discoverable by adding a description. Just start typing...

Location: **lakshmanie** | Refreshed: 3/12/26, 1:05:11 PM | Owner: **Stephanie Bruno**

Tables Filter by keyword

Name	Type	Description
✓ _Measures	Table	
Attendee Count	Measure	
Selected Session	Measure	
Selected Status	Measure	
Session Count	Measure	
Speaker Count	Measure	
Track Count	Measure	